FastGraph: PCA-Subspace Binned k-Nearest-Neighbor Graph Construction for GPU Geometric Deep Learning

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Abstract

Dynamic graph neural networks repeatedly construct k-nearest-neighbor graphs in learned latent spaces, making exact GPU-resident graph construction a central bottleneck for scientific point-cloud workloads. We introduce **FastGraph**, a PyTorch-integrated primitive for exact kNN graph construction in the small-dimensional regimes used by geometric deep learning. The core contribution is a PCA-subspace cell-list formulation: FastGraph learns a compact event-wise binning coordinate system from the leading principal components, uses that subspace for spatial pruning, and evaluates all candidate distances in the original feature space. This removes the dimensional cap of axis-aligned cell lists while preserving exact neighbor sets, extending uniform-grid pruning to the d=4–10 latent spaces used in modern HGCAL reconstruction models. The implementation is GPU-resident end to end and ships with a scripted Python API, a differentiable `GravNetOp` wrapper, and an object-condensation helper kernel for particle reconstruction workflows. On a 5 M-point CMS HGCAL recHit workload, FastGraph builds the exact graph in 7.2 s at d=8, k=40, achieving speedups of over 40× over exact GPU baselines and over 17× over approximate CAGRA while retaining exact graph topology.

Keywords: Graph Neural Networks, k-Nearest Neighbors, GPU Acceleration, CUDA, PCA, Object Condensation, Particle Physics, Differentiable Computing

1. Introduction

Graph neural networks (GNNs) have become a standard substrate for reasoning over irregular relational data, from web-scale recommender systems [1] to particle reconstruction in high-granularity calorimeters [2, 3]. A defining architectural choice in many modern GNNs is that the graph itself is built at training time, from a learned latent embedding, by a k-nearest-neighbor query. The dynamic kNN-graph layer is therefore on the critical path of every forward and backward pass; its construction cost frequently dominates the model's wall-clock budget once the latent dimensionality is anything other than two or three. Importantly, the latent dimensionality in practice is small but not trivially so: the original GravNet layer for HGCAL reconstruction [2] uses d=4 for the learned space in which the kNN is taken, and recent multi-particle and object-condensation variants used in CMS HGCAL reconstruction [3, 4, 5] sit in the d=4–10 range. EdgeConv [6] and ParticleNet [7] are in the same window. This *small-but-not-tiny* regime is precisely where neither generic high-dimensional ANN libraries nor brute-force GPU kNN are an ideal fit: ANN libraries introduce unfavorable overheads, and brute force scales as N^2d regardless of how much spatial structure the latent embedding contains.

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Why exact kNN matters in this regime.. For scientific workloads where the underlying topology must be faithfully recovered, the dynamic-graph layer must also be exact, not just high-recall. In CMS HGCAL shower reconstruction the neighbors that just-barely-fall inside or outside the kth-nearest cut are the structurally informative edges—points that separate one shower from a neighboring shower. An approximate backend at recall=0.9 does not lose 10% of arbitrary edges; it preferentially loses the boundary edges that the GNN most needs, and the loss varies non-deterministically across reruns. This introduces a recall-dependent systematic into training gradients (the dynamic graph fluctuates from epoch to epoch in a way that's a function of the index implementation, not the data) and into inference (downstream physics observables—cluster energy, shower identification—become a function of which neighbors the approximate index happened to return). This is why CMS HGCAL reconstruction pipelines use exact kNN in the dynamic-graph layer even when faster approximate alternatives are available.

The classical low-dimensional accelerator for this regime is the *cell list* [8]: place points in a uniform grid, expand neighboring bins around each query until the best-k radius is certified, and evaluate distances only for candidates in the visited bins. Prior axis-aligned GPU cell lists hit a hard dimensional cap because the bin tensor scales as n_{bins}^d; beyond $d_{\max}$=5, the bin array is impractical and the implementation falls back to brute force. This is precisely the d=6–10 latent-space regime used by modern GNNs.

FastGraph resolves this cap by binning in the top-k principal subspace while retaining full-dimensional distances for candidate evaluation. Since the partial projection $V_k V_k^\top \preceq I_d$ is contractive, projected-space termination remains a valid lower bound on true-space distance, so the algorithm is exact at every input dimension. At d=8, N=5 M, k=40, FastGraph builds the kNN graph in 7.2 s versus 306 s for exact FAISS-GPU, 146 s for cuVS brute force, and 127 s for CAGRA's approximate NN-Descent build. Across d=2–10 at this (N, k) point, geomean speedups are 10.1×, 4.8×, and 4.3× respectively.

The library is shipped as a PyTorch extension: the primary entry point is a `@torch.jit.script`-decorated function callable from inside other scripted modules without graph-capture overhead, and gradients flow through the continuous outputs of the kernel (selected squared distances, query coordinates) while the hard neighbor indices are treated as constants in the backward pass. `GravNetOp` [2] and a GPU-resident object condensation helper [3] are provided as production wrappers; the latter is described in the appendix.

2. Related Work

Exact GPU kNN.. Modern GPUs make the inner-loop distance computation of exact kNN embarrassingly parallel. FAISS [9] provides a highly tuned `IndexFlatL2` exact brute-force GPU index together with the fast k-selection primitives that downstream methods reuse. RAPIDS `cuVS` exposes a comparable exact brute-force GPU implementation that we use as the strongest exact baseline in this work; KeOps [10] provides symbolic matrix reductions with automatic differentiation that can express exact kNN without materializing the full distance matrix. Direct exact GPU kNN-graph construction has also been studied in the multi-GPU and cluster settings [11, 12]. FastGraph is not the first exact GPU kNN library; it specializes the accelerator for the small-d exact graph-construction setting that arises in dynamic GNN layers.

Cell lists and spatial binning.. The cell list is the canonical low-dimensional accelerator: it dates to Verlet's molecular-dynamics neighbor-finding scheme and underpins modern MD libraries such as HOOMD-blue. The construction is the same across communities—uniform grid in d dimensions, sort points by bin, expand a concentric hypercube of bins around each query until the best-k radius is certified—but the choice of d has been constrained in practice to two or three axis-aligned dimensions because the bin tensor scales as n_{bins}^d. qasim2023thesis discusses this cap explicitly in the GNN context; FastGraph's PCA-subspace formulation lifts it.

PCA-projected indexing.. Using a principal-component projection as a coordinate system for spatial indexing has a long history; Sproull's k-d tree refinements [13] already noted that splitting on the direction of maximum variance yields more balanced partitions than splitting on a coordinate axis. PCA trees and random-projection trees use a similar idea for CPU-side approximate nearest-neighbor search. FastGraph's PCA-subspace binning differs in two ways: the projection is partial (top-k axes) rather than full, and the candidate-distance evaluation that follows the pruning step is performed in the *original* d-dimensional space, preserving exactness. The contraction $V_k V_k^\top \preceq I_d$ is what makes that combination valid (Section 3).

Three-dimensional spatial-acceleration kNN.. A separate line of work targets exact GPU kNN through dedicated spatial-acceleration structures that exploit the GPU's ray-tracing hardware or BVH machinery. RTNN [14], RT-kNNS Unbound [15], and Arkade [16] retarget the OptiX BVH unit for neighbor search and report large speedups over shader-core baselines in 3D. LBVH-kNN [17] (`cupy-knn`) attains similar gains via a left-balanced LBVH built without dedicated RT hardware. CLOVER [18] introduces a GPU-native, Voronoi spatio-graph approach to exact kNN and reports 4× over an "optimized grid" baseline and 230× over FAISS on SIFT-like data. All of these methods are fundamentally locked to two or three input dimensions: Arkade notes that RT cores build BVHs only on three-dimensional data, CLOVER's reference implementation asserts `D == 3`, and LBVH-kNN is evaluated exclusively in 3D. We include a direct head-to-head against CLOVER at d=3 in Section 5.6, but these methods do not cover the d=5–10 latent-space workload targeted here.

Approximate GPU kNN.. A parallel body of work has produced GPU-resident approximate nearest neighbor (ANN) systems. SONG [19] decomposes graph traversal into GPU-friendly stages; GGNN [20] builds a hierarchical kNN graph for index construction and search; CAGRA [21], distributed as part of RAPIDS `cuVS`, builds a high-quality graph index with an IVF-PQ or NN-Descent initialization. ANN-Descent variants have also been adapted to reduce memory traffic during construction [22]. These systems target high-throughput vector retrieval at recall $\lesssim 0.9$; FastGraph occupies the orthogonal exact, small-d, graph-construction regime where the cost of an approximate edge is not borne by a single retrieval call but accumulates across every training event. We benchmark CAGRA and GGNN as approximate comparators (Section 5.7) but do not include CPU-only ANN systems (HNSW, ScaNN, Annoy) [23, 24, 25] in the timing tables: the workload here is GPU-resident end-to-end and a CPU-side detour would not be a fair comparator.

Dynamic latent-graph models in geometric deep learning.. Dynamic kNN-graph layers in learned latent spaces are a recurring ingredient in geometric deep learning. DGCNN/EdgeConv recomputes the graph in feature space at each layer [6]; ParticleNet adapts the same idea to jet tagging [7]; GravNet/GarNet layers [2] and the object condensation loss [3] are deployed for HGCAL reconstruction [4, 5]; the HEP.TrkX/Exa.TrkX line of work uses learned-embedding graphs for tracking [26, 27, 28]. FastGraph is best framed as an exact substrate for the kNN-graph layer inside such models—not a new model class.

3. The PCA-Subspace Binned kNN Algorithm

3.1. Phase 1: Subspace estimation

Given the input coordinates $X \in \mathbb{R}^{N\times d}$ for the current event (or mini-batch), FastGraph computes the top-k principal axes via a partial eigen- decomposition of the centered covariance $X^\top X$. The number of binning axes k is a user-facing hyperparameter; we default to $k = 3$ for the HGCAL workloads in this paper, justified empirically in the ablation of Section 5.5. The resulting projection matrix $V_k \in \mathbb{R}^{d\times k}$ satisfies $V_k^\top V_k = I_k$ and $V_k V_k^\top \preceq I_d$. The projected coordinates used for binning are $Y = XV_k$. PCA estimation is performed once per event on the GPU using the underlying BLAS implementation and contributes negligibly to the wall-clock budget relative to the per-query search.

Short-circuit at $d \le k$.. When the input dimensionality d does not exceed the binning dimensionality k, FastGraph skips PCA estimation entirely and binds the binning axes to the raw input axes; the projection V_k reduces to the identity (up to permutation), $Y = X$, and the algorithm behaves exactly as a classical axis-aligned cell list. With the default $k = 3$, the $d \in \{2, 3\}$ gains therefore come from the underlying cell-list machinery; the PCA contribution begins at $d > k$, which covers the d=4–10 HGCAL latent-space range.

3.2. Phase 2: Binning in the projected subspace

We partition Y with a uniform grid of width w along each of the k projected axes. The per-axis division count is set heuristically to balance bin density against neighborhood reach,

$$n_{\text{bins}} = \left(\frac{32\, n_{\text{elems}}}{K}\right)^{\frac{1}{k}},$$

where n_{elems} is the per-row-split point count, K is the requested neighborhood size, and the result is clamped to $[5, 30]$. Points are sorted by bin index, and cumulative offsets are stored so that each bin can be addressed as a contiguous slice of the sorted point array. Row splits, which delineate the per-event boundaries inside a batched tensor, are honored by the binning kernel so that no query bin spans events.

3.3. Phase 3: Per-query search with full-dim distances

For each query x_q, the search starts in the bin containing $y_q = x_q V_k$ and expands through successive shells of k-D hypercube bins. For every candidate u in a visited bin the *full-dimensional* squared Euclidean distance $\|x_q - x_u\|^2 = \sum_{i=1}^{d} (x_{q,i} - x_{u,i})^2$ is computed (not the k-dimensional projected distance). The current best neighbor radius r_K and the heap of nearest-K candidates are maintained incrementally.

3.4. Why this is exact: the contraction argument

For any dataset size N, ambient dimension d, projection dimension $k \leq d$, and neighbor count K, consider the standard cube-radius termination test applied to bins in the projected subspace while all candidate distances are evaluated in the full input space. The exactness of this test rests on a single observation about the partial projection. For any two points $x_a, x_b \in \mathbb{R}^d$,

$$\|y_a - y_b\|^2 = (x_a - x_b)^\top V_k V_k^\top (x_a - x_b) \leq \|x_a - x_b\|^2,$$

because $V_k V_k^\top$ has eigenvalues in $\{0, 1\}$ and is therefore $\preceq I_d$. Equivalently, the projection is contractive: the projected distance never overestimates the true distance.

After visiting all bins within the current hypercube shell of radius s, any unvisited bin lies at projected distance at least $w \cdot s$ from y_q, so any unvisited point satisfies $\|y_q - y_u\|^2 \geq (w \cdot s)^2$. Combined with the contraction above, $\|x_q - x_u\|^2 \geq (w \cdot s)^2$, so once the heap has filled and $(w \cdot s)^2 > r_K^2$, no unseen point can be closer in the true distance than the current K-th neighbor. The algorithm may safely terminate; the returned K neighbors are exact. We additionally verify exactness empirically by comparing against a PyTorch brute-force reference at the tractable scale, obtaining bit-identical neighbor indices and zero distance deltas across all spot-checked configurations.

3.5. Binstepper

The `binstepper` is the small bookkeeping utility that enumerates the surface of the s-th hypercube shell around the query bin. It linearises each shell into a flat sequence, converts each step back into per-dimension offsets, discards interior cells, and yields the next surface bin's flat index. The stepper carries no algorithmic role; the kNN guarantee lives in the outer kernel of Algorithm 1.

3.6. Pseudocode

Algorithm 1 makes the projected-vs-original-space distinction explicit: the bin indices (`binIdx`) and the stepper operate on y-space; the distance function `calcDist` operates on the original x-space coordinates; the termination test references the projected-space cube radius; the contraction argument certifies the result.

Algorithm 1: PCA-Subspace Binned-Select-kNN: bins live in projected y-space; distances are computed in original x-space.

Input: `coords`$[n_v][n_c]$ (original-space X), V_k, `binIdx`$[n_v]$ (projected-space bins), `binOffsets`$[n_v][n_b{+}1]$, `binBounds`$[n_B]$, `binCounts`$[n_b]$, `binWidths`$[n_b]$, n_v, K, n_c, n_b, n_B

Output: `neighbors`$[n_v][K]$ indices, r_K^2 values

for each $v \in [0, n_v)$ ***in parallel*** **do**
  `neighbors[v][0]`$\leftarrow (v, 0)$; `filled`$\leftarrow 1$; $(\texttt{maxIdx}, \texttt{maxD2}) \leftarrow (0, 0)$
  `totalBins` $\leftarrow \prod_{d=0}^{n_b-1}$ `binCounts`$[d]$, `gOff` $\leftarrow$ `totalBins` $\cdot \lfloor$`binIdx`$[v]$/`totalBins`$\rfloor$
  `step`$\leftarrow$`BinStepper(binCounts, binOffsets[v][1..`n_b`])` `// operates in projected y-space`
  `cont`$\leftarrow$ true; $s \leftarrow 0$ `// s: current cube-shell radius (projected space)`
  while *cont* **do**
    `step.setDistance(`s`)`; `cont`$\leftarrow$ false
    while (`off` $\leftarrow$ `step.step()`) $\neq -1$ **do**
      `idx` $\leftarrow$ `off` + `gOff`
      if *idx* < 0 ***or*** *idx* $\geq n_B - 1$ **then**
        continue
      $(s_b, e_b) \leftarrow$ (`binBounds[idx]`, `binBounds[idx+1]`)
      if $s_b = e_b$ **then**
        `cont`$\leftarrow$ true; **continue**
      for $u \in [s_b, e_b)$ **do**
        if $u{=}v$ **then**
          continue
        `d2` $\leftarrow$ `calcDist(`v, u`)` `// full d-dim` $\sum_i (x_{v,i} - x_{u,i})^2$`, NOT projected`
        if *filled* $< K$ **then**
          `neighbors[v][filled]`$\leftarrow (u,$ `d2`$)$; **if** *d2* $>$ *maxD2* **then**
            $(\texttt{maxIdx}, \texttt{maxD2}) \leftarrow (\texttt{filled}, \texttt{d2})$
          ; `filled++`
        else if *d2* $<$ *maxD2* **then**
          `neighbors[v][maxIdx]`$\leftarrow (u,$ `d2`$)$; $(\texttt{maxIdx}, \texttt{maxD2}) \leftarrow$ `findMaxDist(neighbors[`v`])`
      `cont`$\leftarrow$ true
    if *filled*$=K$ ***and*** $(\mathit{binWidths}[0] \cdot s)^2 >$ *maxD2* **then**
      break
    s++

3.7. Complexity

Let $B = n_{\text{bins}}^k$ be the size of the bin grid. The PCA projection reduces this from the n_{bins}^d of an axis-aligned scheme to a quantity that depends only on the binning subspace size k, *independently of the input dimension* d. This is what removes the small-d cap that limited prior cell-list GPU kNN implementations.

Setup. Subspace estimation runs a partial eigen-decomposition of $X^\top X$ at $O(Nd^2 + d^3)$, dominated in our regime by the Nd^2 Gram accumulation. Binning, sorting, and offset construction take $O(N \log N + B)$.

Memory. Auxiliary memory is $O(N(d + k + K) + B)$: sorted coordinates, projected coordinates, permutation, neighbor indices, and bin offsets.

Per-query search. Under roughly uniform projected-space density, each query examines $O(K)$ useful candidates before certifying r_K; each candidate costs $O(d)$ for the full-dim distance and at most $O(K)$ for the heap update, giving expected $O(K(d + K))$ per query. The worst case is the degenerate all-points-in-one-bin configuration, which falls back to the brute-force $O(N(d + K))$.

Parallel cost. With P CUDA threads,

$$\frac{N}{P} \cdot O(K(d+K)) \;+\; O\Big(Nd^2 + N\log N + B\Big).$$

3.8. Gradient flow and PyTorch integration

FastGraph exposes gradients through the continuous outputs of the kernel—the selected squared distances and the input coordinates— while treating the discrete neighbor indices and the projection matrix V_k as constants in the backward pass. Concretely, the forward returns (idx, dist^2) and the backward applies the chain rule with respect to dist^2 and the input coordinates, yielding a subgradient of the piecewise-smooth loss induced by the hard neighbor selection. This is the same gradient model used by any fixed-index distance layer (e.g., `torch.cdist` followed by `topk`) and integrates naturally into existing GNN autograd graphs. Treating V_k as fixed per forward+backward pass keeps the gradients deterministic within a step; the projection is refreshed at the next forward.

The user-facing entry point is decorated with `@torch.jit.script`, so it can be called from within other scripted modules without graph-capture overhead. A minimal usage example:

```
from fastgraphcompute import binned_select_knn_pca  # @torch.jit.script decorated

# Direct call (eager mode)
idx, dist2 = binned_select_knn_pca(K=40, coords=x, row_splits=rs,
                                   max_bin_dims=3)
# idx:   (N, K) int64 neighbor indices (constant in backward)
# dist2: (N, K) float32 squared distances (differentiable)
loss = model(dist2).sum()
loss.backward()  # gradients flow through dist2 to x

# Inside a scripted module
@torch.jit.script
def build_graph(x: torch.Tensor, rs: torch.Tensor) -> torch.Tensor:
    idx, dist2 = binned_select_knn_pca(40, x, rs, 3)
    return dist2
```

Listing 1: Minimal FastGraph usage inside a `torch.jit.script` module.

4. Experimental Setup

All GPU experiments use a single NVIDIA A100 80 GB PCIe with CUDA 12.1 and PyTorch 2.5. Compared backends are exact FAISS-GPU `IndexFlatL2` [9], cuVS brute force (also exact), CAGRA via cuVS [21] (approximate), and GGNN [20] (approximate); FastGraph is compiled from the public release.

Timing protocol.. Timing runs use a single A100 allocation through `-gres=mps:100`. All timed regions are bracketed by `torch.cuda.synchronize()` calls on the active CUDA stream, after warm-up for memory allocation and JIT compilation. Figures and scalar comparisons report medians across successful repetitions; error bars show the interquartile range when repeated measurements are available.

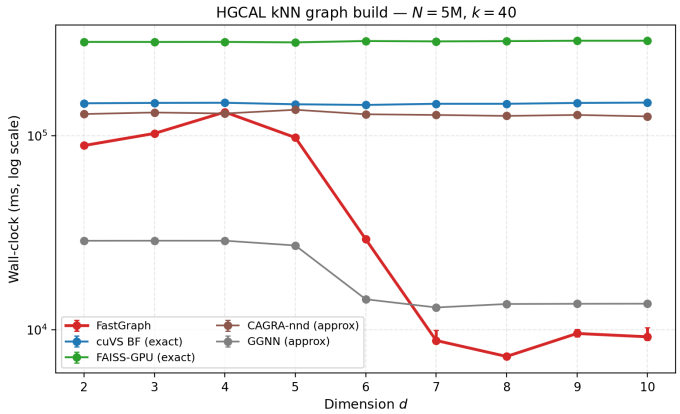
CAGRA build path.. CAGRA's default IVF-PQ build path fails on HGCAL data at $d \geq 5$ with a RAFT assertion about invalid or duplicated neighbor nodes. The failure is consistent with PQ codebook collapse on heterogeneous-scale features: energy, geometry, time, and layer index occupy very different numeric ranges. We therefore use CAGRA's `build_algo="nn_descent"` path throughout the paper; it avoids the PQ initialization, succeeds across the full d=2–10 sweep, and is the stronger CAGRA baseline used in all plots and tables.

The `max_bin_dims` hyperparameter.. The binning subspace size k is exposed as `max_bin_dims`. The current CUDA implementation provides compile-time kernel template instantiations for $k \in \{2, 3, 4, 5, 6\}$; we use $k = 3$ as the default for HGCAL and ablate the choice in Section 5.5. The kernel block size is fixed at 512 threads, which leaves enough register budget for the $k \leq 6$ templates; raising k further would require a smaller launch configuration. The empirical optimum on HGCAL data is well inside the supported range, so this is not a binding limitation in practice.

Benchmark data. The CMS HGCAL recHit feature dataset comprises 1.16 billion hits concatenated across all events with up to ten spatial and energy features per hit. Synthetic comparisons use isotropic Gaussian $\mathcal{N}(0, I_d)$ samples; this matches our actual benchmark protocol and isolates the effect of dimensionality from the heterogeneous-scale pathology of HGCAL.

5. Performance Comparisons

5.1. Headline benchmark: Dimensional scaling on detector data



d	FAISS	cuVS	CAGRA	GGNN
2	3.42×	1.65×	1.45×	0.32×
3	2.96×	1.43×	1.28×	0.28×
4	2.30×	1.12×	0.98×	0.22×
5	3.09×	1.48×	1.39×	0.28×
6	10.51×	4.92×	4.40×	0.49×
7	35.23×	16.77×	14.53×	1.48×
8	42.46×	20.15×	17.36×	1.86×
9	32.68×	15.60×	13.37×	1.42×
10	29.83×	14.31×	13.67×	1.48×
geo.	10.1×	4.8×	4.3×	0.63×

Figure 1: Headline benchmark on detector data at N=5 M, k=40. Left: dimensional scaling of wall-clock time. FastGraph is the fastest exact GPU method across the full d=2–10 range. Right: speedups of FastGraph over each backend at the same operating point. FAISS and cuVS BF are exact; CAGRA-nnd and GGNN are approximate default-recall baselines, and GGNN's timing should be read with its substantially lower recall in this regime (Section 5.7).

The right-hand table in Figure 1 reports the cross-method speedups of FastGraph at the headline operating point. FastGraph is the fastest *exact* GPU kNN method at every dimension we measure and has a 4.3× geomean advantage over the approximate CAGRA-NN-Descent build; the d=4 CAGRA cell is an essentially tied 0.98× case. The GGNN approximate baseline does win on raw wall-clock at low d, but at recall well under 1.0 in matched-recall evaluation (Section 5.7)—a tradeoff the exact-neighbor regime targeted by this paper cannot accept. The relative advantage is widest in the d=7–10 regime where dense exact baselines pay the full quadratic cost and the approximate baselines have to retain enough graph quality to remain useful. Geomean across d=2–10 at this operating point is 10.1× over FAISS, 4.8× over cuVS BF (the strongest exact GPU baseline), and 4.3× over CAGRA-NN-Descent. Section 5.5 separately reports the internal axis-aligned ablation that motivated the PCA-subspace design; we keep that comparison out of the headline table so the public method identity is simply FastGraph.

The qualitative shape of the curve is determined by the regime transition at d=5–6. Below this transition, all GPU methods are within a small constant factor and the binning advantage is modest; above it, FAISS/cuVS BF (which scale as $N^2 d$) both grow rapidly with d, while FastGraph retains a useful pruning structure through PCA-subspace binning.

5.2. Dataset-size scaling

Figure 2 reports dataset-size scaling at fixed k=40 for d=3 and d=8; the same ordering persists as N grows, with stronger separation in the PCA-favorable regime.

Figure 3 checks that the headline dimensional trend is not an artifact of the single k=40 operating point. Across $k \in \{10, 40, 100\}$, the low-dimensional constant-factor regime and the larger $d \geq 6$ speedups remain visible.

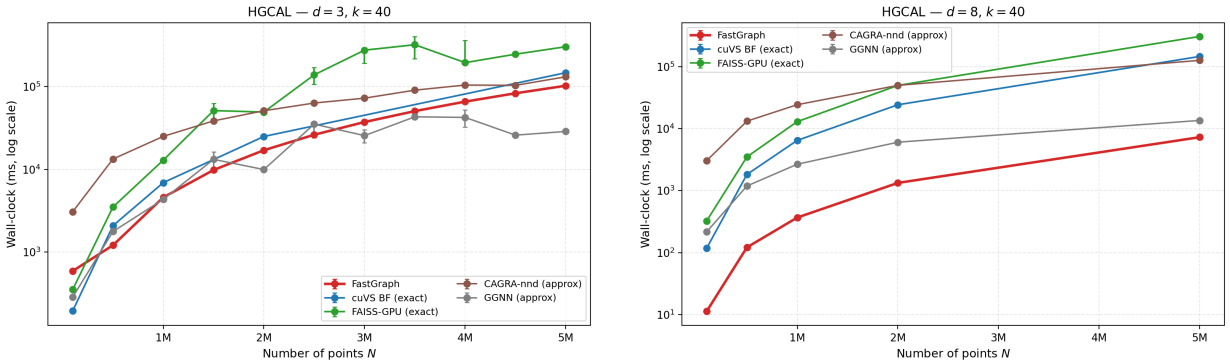


Figure 2: Dataset-size scaling on HGCAL at k=40 for d=3 (left) and d=8 (right). FastGraph maintains a speed advantage over exact baselines from small samples through N=5 M, with the gap widening in the higher-dimensional PCA-favorable regime.

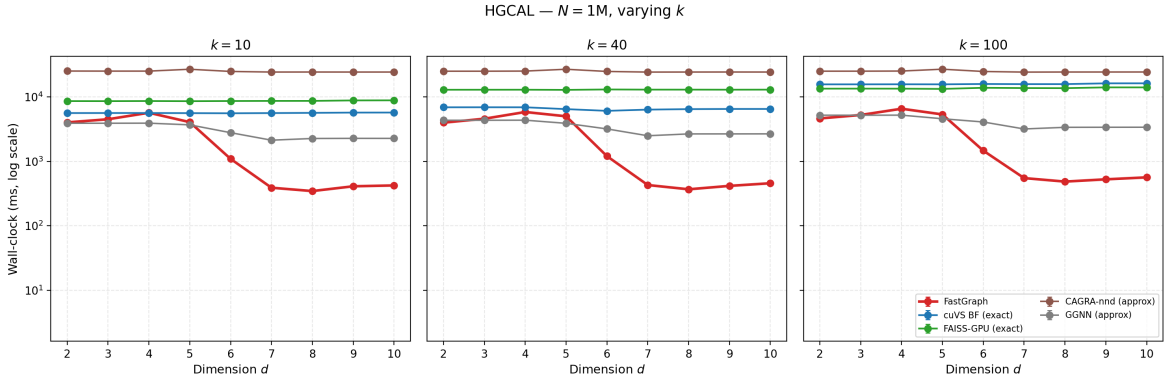


Figure 3: Speedup at N=1 M for $k \in \{10, 40, 100\}$ across the full dimensional range. The qualitative pattern of Figure 1 is preserved across k.

5.3. Controlled synthetic benchmarks

On isotropic Gaussian $\mathcal{N}(0, I_d)$ data, FastGraph wins decisively at low dimensions, while dense brute-force GPU matmul (cuVS BF) becomes competitive as d grows. This is consistent with the algorithmic interpretation: the PCA projection contributes nothing on isotropic data — every direction carries equal variance, so the principal axes are an arbitrary unitary rotation of the input and the wrapper degrades to axis-aligned binning. At low d, cell-list pruning still dominates the dense $O(N^2 d)$ matmul of the exact GPU baselines, giving FastGraph a 55× speedup over cuVS BF at d=3 and a 4.8× speedup at d=5 (Table 1). As d grows the number of neighboring cells grows combinatorially, the binning savings collapse, and dense matmul on a well-tuned BLAS path is close to optimal on isotropic data. Figure 4 shows the full dimensional sweep and two dataset-size scans at d=3 and d=8.

Table 1: FastGraph speedup over GPU baselines on synthetic Gaussian $\mathcal{N}(0, I_d)$ data at N=10^6, k=40 (median of 3 reps, Binary A). The HGCAL detector-data wins at $d \geq 6$ (Figure 1) are not reproduced on isotropic data, isolating them to the anisotropic structure of detector features rather than a generic property of cell-list binning.

d	vs cuVS BF	vs FAISS-GPU	vs CAGRA-nnd
3	55.2×	108.6×	205.8×
5	4.8×	9.4×	18.0×
7	0.8×	1.7×	3.2×

The implication for the paper's main claim is that the HGCAL detector wins at d=6–10 (Section 5.1) come from PCA picking up anisotropic structure that does not exist in isotropic synthetic data, not from cell-list binning alone.

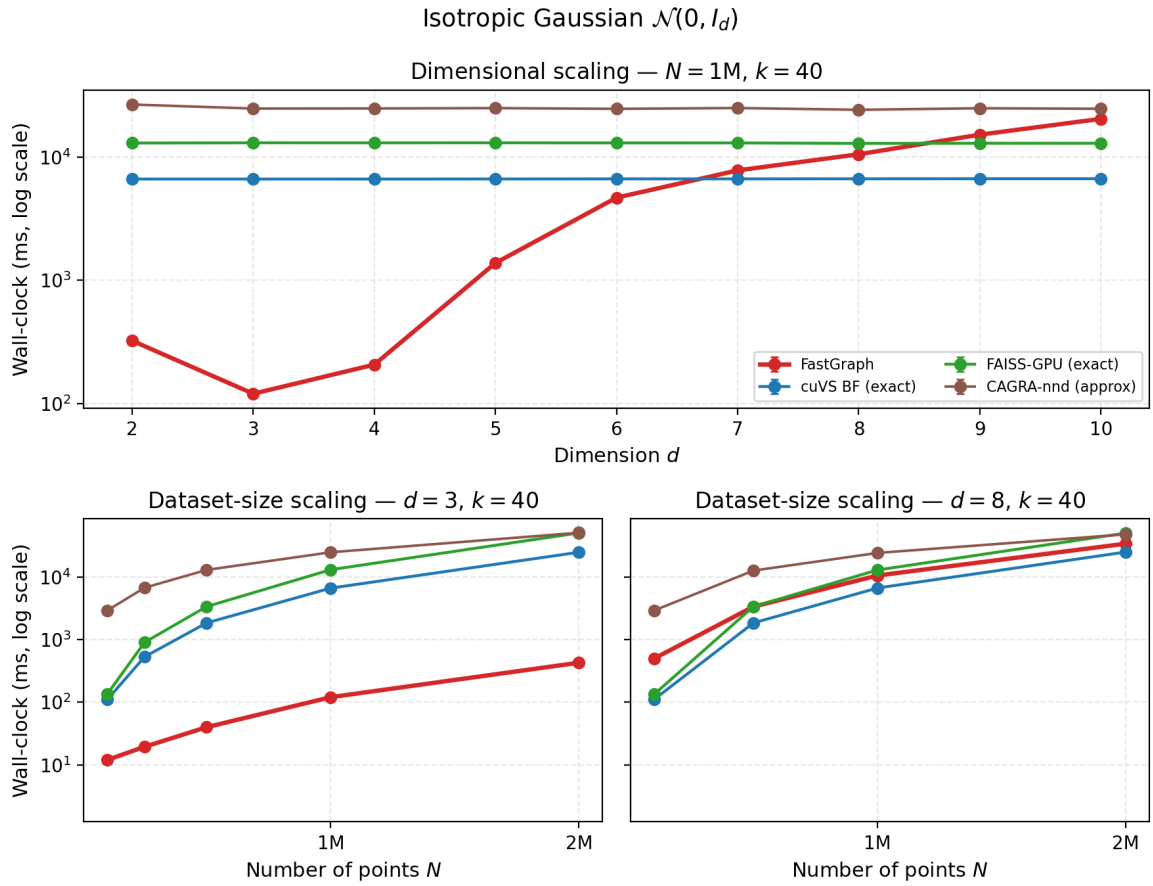


Figure 4: Controlled synthetic Gaussian benchmarks. Top: dimensional scaling at N=1 M, k=40. Bottom: dataset-size scaling at d=3 and d=8, k=40.

5.4. *Memory footprint*

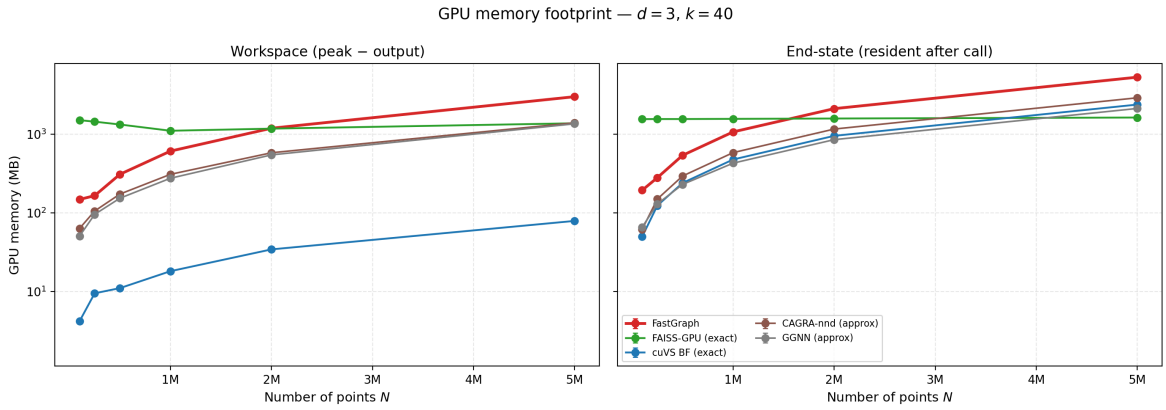


Figure 5: GPU memory footprint per backend at d=3, k=40, measured by a 100 Hz NVML polling thread that captures the true peak during the call. *Left:* workspace, defined as peak − (N×k) output bytes, i.e. everything the algorithm allocates beyond the (idx, dist) output tensors; this isolates temporary algorithm workspace from the common output allocation. *Right:* end-state, the resident NVML reading after the call synchronises while the caller still holds the output tensors.

Workspace and resident memory.. FastGraph allocates more workspace than the dense GPU baselines: at N=5 M, d=3, k=40 it uses about 3.0 GB versus 1.4 GB for FAISS-GPU, CAGRA-nn-descent, and GGNN, and ~ 80 MB for cuVS BF; the corresponding FastGraph workspace at d=8 is ~ 3.3 GB. This cost comes from the projected coordinate array, bin assignments and boundaries, sort/remap intermediates, and tensors required by the differentiable distance outputs. In resident end-state memory, the gap is smaller: FastGraph holds ~ 5.3 GB versus 1.6–2.9 GB for the other backends, because the common $N\times k$ output tensors dominate resident memory after temporary workspace is released.

Approximate-backend interpretation.. The plot reports each backend at the configuration used elsewhere in this paper. FAISS-GPU `IndexFlatL2` is exact; cuVS BF is exact. CAGRA-nn-descent and GGNN are approximate at their default quality settings; their memory advantage is partly inherited from running at lower effective recall (see Section 5.7). A matched-recall memory comparison would require either raising their quality parameters until recall hits a target or downsampling their internal graphs, neither of which has a canonical setting; we therefore report the memory each backend uses *as benchmarked*, with the understanding that the approximate backends are not bound to FastGraph's exact-recall constraint.

Measurement methodology.. Memory is measured with a background NVML poller running at ~ 100 Hz during the timed region. The reported workspace is the observed peak minus the backend's output-tensor footprint, while the end-state value is the resident allocation after synchronization with the output tensors still live.

5.5. Binning dimensionality ablation

The binning subspace size k (`max_bin_dims`) is the most important FastGraph hyperparameter. We set the default to k=3 because it is the fastest PCA-subspace setting at the reference cell d=8, N=5 M, k_{nn}=40. Table 2 compares this choice with the axis-aligned cell list over the released kernel surface $k \in \{2, 3, 4, 5\}$, using medians over three repetitions.

Table 2: Effect of the binning subspace size k on FastGraph wall-clock at d=8, N=5 M, k_{nn}=40, median of three repetitions. The axis-aligned variant's optimum within the released kernel surface is k=2; the PCA-subspace variant's runtime optimum, and therefore the default used in the paper, is k=3.

k (bin dims)	axis (s)	PCA (s)	PCA/axis speedup
2	199.37	11.39	17.5×
3	204.65	**7.48**	**27.4**×
4	353.78	11.58	30.6×
5	242.90	28.10	8.6×

Both variants are non-monotonic in k, but their optima differ: the axis-aligned cell list is fastest at k=2, while the PCA-subspace variant is fastest at k=3. The largest matched-k speedup occurs at k=4, but k=3 is the setting used by FastGraph because it minimizes the PCA-subspace wall-clock. Across the sweep, PCA-subspace binning is 8.6–30.6× faster at matched k, and comparing each variant at its own optimum gives a 26.6× speedup.

We expose k as a user-facing hyperparameter rather than hard-coding it: the optimum is data-dependent (a dataset with intrinsic dimensionality near 2–3 prefers small k; one with variance spread more uniformly prefers k=4 or 5), and the same kernel binary supports all four instantiations. The default k=3 is HGCAL-tuned and should be revalidated when deploying FastGraph on datasets with different intrinsic dimension.

5.6. Three-dimensional baseline: CLOVER head-to-head

CLOVER [18] is the strongest among the 3D-only exact-GPU kNN methods surveyed in Section 2 and the only published method on this list whose absolute timings exceed FastGraph's on real data. This comparison does not weaken the paper's main claim: CLOVER is restricted to raw 3D point clouds, whereas FastGraph targets exact kNN in learned latent spaces with d=4–10.

The result splits along the structure of the input distribution. On uniform-cube data — the regime where a uniform-grid cell list is near-optimal — FastGraph is 5–8× faster than CLOVER hubs. On isotropic Gaussian data the two methods are within roughly a factor of two. On anisotropic HGCAL recHits, CLOVER's hubs index is much faster at

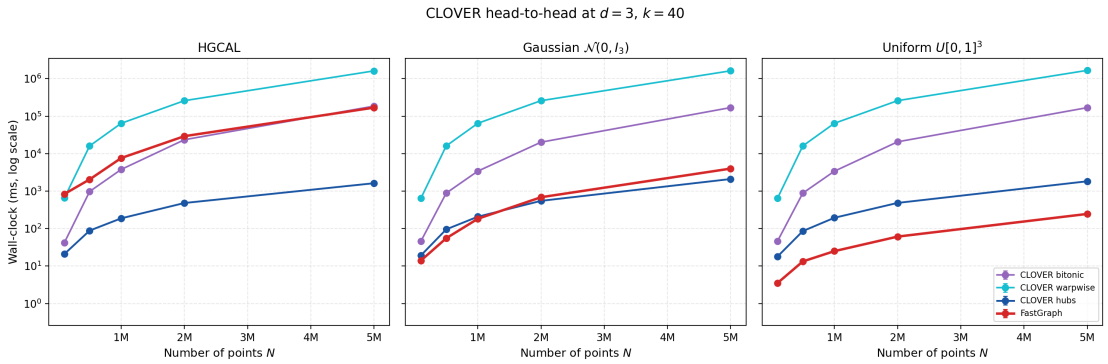


Figure 6: FastGraph vs CLOVER at d=3, k=40, across three data distributions: HGCAL recHits (left), synthetic Gaussian $\mathcal{N}(0, I_3)$ (center), synthetic uniform on $[0, 1]^3$ (right). CLOVER's reference implementation is hardcoded to D=3 at multiple sites (`assert(D == 3)` in `src/linear-scans.cu`; `auto const dim = 3u` in `include/spatial.cuh`), so d=3 is the only dimensionality at which a direct comparison is possible.

d=3 because it reorganises queries around point density rather than around a uniform Euclidean grid. This is exactly the native regime of a 3D spatial graph, but it is not the regime targeted by FastGraph's PCA-subspace contribution.

The deployment regime for HGCAL graph-construction GNNs is the *learned latent space* of the trained network, not the raw 3D detector coordinates. Modern dynamic-graph layers used in HEP reconstruction operate at $d = 4$–10 in this latent space: GravNet uses $d = 4$ in its original formulation [2] and $d \in \{6, 8, 10\}$ in recent HGCAL multi-particle variants [4, 5]; EdgeConv-derived layers [6] and object condensation pipelines [3] expose similar ranges. CLOVER, LBVH-kNN, and the RT-core methods are unavailable by construction in this regime: they are hardware-locked or implementation-locked to $D = 3$ and have no defined extension to $d \geq 4$.

5.7. Recall evaluation

To verify FastGraph's exactness we compare against a brute-force reference on HGCAL data up to N=5×10^5. Because many recHits are spatially coincident, neighbor indices are not unique; we therefore use *distance-based recall*, counting a predicted neighbor as correct when its squared distance is no larger than the k-th reference distance plus a small tolerance. FastGraph and its reference use element-wise squared ℓ_2 distances, which avoid the cancellation that can affect coincident-hit cases under the BLAS expansion. FAISS's `IndexFlatL2` is evaluated against a BLAS-consistent reference so exact methods are not penalized for kernel-specific floating-point differences.

Under this metric FastGraph achieves recall = 1.0 across every tested configuration ($d \in \{2, \ldots, 10\}$, $k_{\text{nn}} \in \{10, 40, 100\}$, N up to 5×10^5). The approximate baselines, even at their highest-quality parameter settings, do not reach exact recall. Table 3 summarises.

Table 3: Distance-based recall on HGCAL data at the highest-quality parameter setting for each backend (`itopk_size`= 512 for CAGRA-NN-Descent, $\tau_{\text{query}} = 0.8$ for GGNN), under the kernel-matched evaluation described in the text. The representative cell is d=3, N=5×10^5, k_{nn}=40. "d=5 collapse" marks the configuration where GGNN's recall falls to ~ 0.03 even at its highest-quality knob.

Method	mean over all cells	representative cell	d=5 collapse
FastGraph (exact)	1.000	1.000	—
CAGRA-NN-Descent	0.732	0.818	—
GGNN	0.666	0.410	0.027

Two points are most relevant. First, neither approximate method achieves 99% distance-based recall at $N \geq 5 \times 10^5$ in the tested cells; raising their search-quality settings to close this gap removes their wall-clock advantage over FastGraph. Second, GGNN's highest-quality setting drops sharply at d=5 before recovering at higher dimensions, so we report both the mean and a representative deployment cell rather than relying on the aggregate alone.

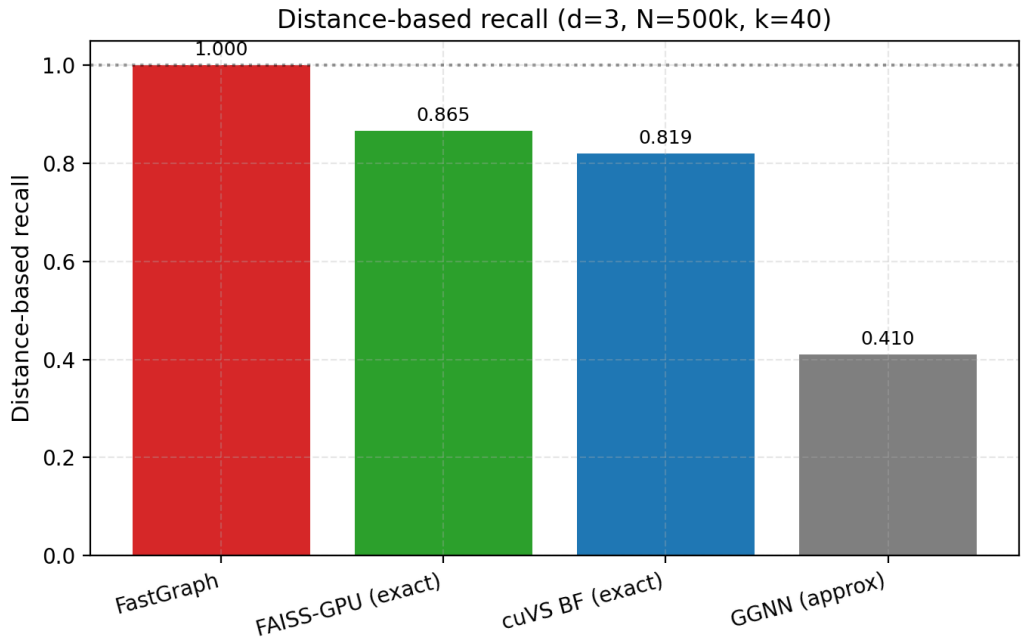


Figure 7: Distance-based recall for FastGraph across detector-derived configurations. The contraction argument plus the bit-identical comparison against brute force at small N together support recall = 1.0 across the full sweep.

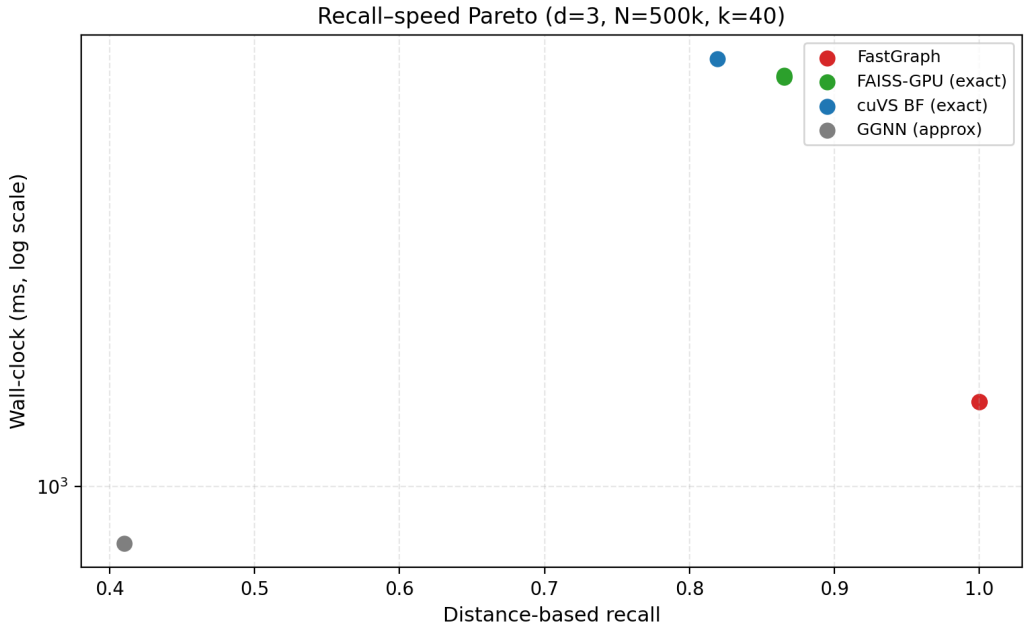


Figure 8: Recall-speed Pareto. FastGraph occupies the exact / fast region; approximate methods are only competitive when lower recall is acceptable.

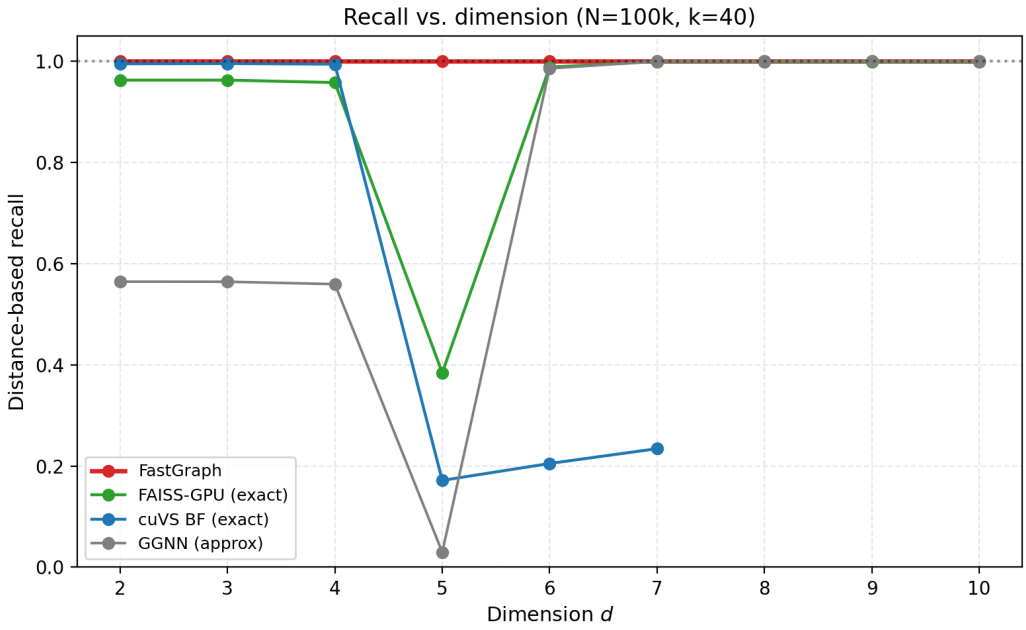


Figure 9: Distance-based recall versus dimension at the best-quality parameter setting (N=10^5, k=40). FastGraph holds recall = 1.0 across every dimension; CAGRA and GGNN degrade above d=3 and neither reaches 99 %.

5.8. Dynamic graph construction in GNNs

GravNet-style layers [2] construct a fresh kNN graph in a learned latent space at every forward pass, so the kNN substrate is on the gradient path of every training step. FastGraph integrates into this setting through `GravNetOp`, a PyTorch module that wraps coordinate transformation, kNN graph construction (via the PCA-subspace primitive of Section 3), and message passing into a single autograd-compatible operation. Gradients flow through the selected distances and the input coordinates; the discrete neighbor indices and the per-event projection V_k are held fixed across the forward-backward pair so that the gradients are deterministic within a step. Object condensation [3], which is the loss used together with GravNet for HGCAL reconstruction, is supported through a separate GPU-resident helper kernel described in the appendix.

6. Limitations

Resolved axis-aligned limitations.. The $d_{\max}$=5 binning cap inherited from axis-aligned cell-list implementations is removed by the PCA-subspace formulation: the projected grid is n_{bins}^k in the binning subspace size k, independent of the input dimensionality d. Per-column scale heterogeneity is also absorbed into the variance ordering of the principal components, so the binning structure no longer has to be told which input axes are large.

Data distribution still matters.. The PCA contribution's effectiveness is a function of how much pairwise-distance variance the leading components actually capture. On strongly anisotropic data—HGCAL recHits, anything with a few dominant axes—FastGraph's PCA-subspace binning is substantially faster than axis-aligned cell lists (Section 5.5, geomean speedups in Section 5.1). On isotropic data, the projection is essentially a unitary rotation and FastGraph degrades gracefully toward ordinary cell-list behavior; in that regime FastGraph outperforms dense GPU brute force only while the cell-list pruning advantage outweighs the constant factor of an optimized BLAS `gemm`. Section 5.3 documents this on Gaussian $\mathcal{N}(0, I_d)$ inputs and is the reason we do not claim FastGraph is the fastest GPU exact kNN on arbitrary distributions; we claim it is the fastest GPU exact kNN in the operating regime where the leading principal components carry real structural information, which is the regime modern learned-latent dynamic graphs are designed to live in.

Remaining limitations.. PCA estimation itself adds an $O(Nd^2)$ term that is amortised over the kNN query in the regimes we benchmark, but at very small N the projection cost becomes a non-negligible fraction of the total wall-clock; for tiny per-event point counts an axis-aligned variant remains preferable. The current implementation targets a single GPU; multi-GPU or distributed kNN graph construction is not supported. The discrete neighbor indices are not differentiable; FastGraph's autograd contract is over the continuous distances and coordinates only, matching every other hard-selection kNN layer. Among the GPU baselines, approximate methods remain faster than FastGraph at default low-recall settings; FastGraph is the right choice only when exact graph topology is required.

Why exactness matters for the downstream GNN.. The case for an exact kNN backend over a recall-tuned approximate one is not a raw-recall argument but a sensitivity-of-downstream argument. Adversarial-robustness studies of graph neural networks show that node predictions can shift substantially under small, structure-preserving edge perturbations to the input graph [29]; the same literature also documents that systematic reweighting of the neighbourhood—e.g., replacing direct neighbours with a diffusion-smoothed neighbour set—measurably alters downstream accuracy [30]. The neighbour-set mismatches injected by an approximate kNN are exactly this kind of perturbation, distributed non-uniformly across the event and dependent on the build-side index hyperparameters. We do not claim the HGCAL reconstruction targets are unusually sensitive to approximate-kNN edge noise—that would require a dedicated model study we have not done—but the more conservative choice for a physics application that demands reproducible, systematics-controlled inference is the exact graph, particularly when an exact backend is competitive on wall-clock in the relevant operating regime (Section 5.1).

CAGRA on heterogeneous data.. CAGRA's default IVF-PQ build path fails on HGCAL data at $d \geq 5$ (Section 4); the NN-Descent fallback succeeds across the full dimensional sweep but is 17.36× slower than FastGraph at the d=8 reference cell. We report the NN-Descent numbers throughout. This is a finding about CAGRA on heterogeneous-scale features, not a generic CAGRA limitation; practitioners deploying CAGRA on similarly heterogeneous physics datasets should be aware of the failure mode and the appropriate build-path mitigation.

7. Data and Code Availability Statement

The FastGraph library, including the PCA-subspace extension described in this paper, is released at `https://github.com/AgarwalAarush/FastGraphCompute` (fork of the upstream library at `https://github.com/jkiesele/FastGraphCompute`). The exact source used to collect every headline timing in this paper is the `paper-release` branch built against the CUDA 12.1 toolchain. The released kernel surface covers binning subspace size $k \in \{2, 3, 4, 5\}$. The benchmarking harness and runner scripts used to produce the figures live in a companion repository, `https://github.com/AgarwalAarush/fgc-performance`. The CLOVER head-to-head used a fork of `https://github.com/ampslab/clover-knn` at `https://github.com/AgarwalAarush/clover-knn` with local source modifications (`k_values` aligned to the paper's k=40, the hardcoded synthetic-N template chain in `main()` disabled in favor of the mesh path, and the build retargeted to `sm_80` for A100).

The detector-derived benchmarks use CMS HGCAL simulation data (recHit feature vectors) accessed via an internal collaboration dataset of 1.16 billion calorimeter hits with up to ten spatial and energy features per hit. The data itself is internal to the CMS collaboration; synthetic Gaussian and uniform datasets used in Sections 5.3 and 5.6 are generated on the fly from fixed random seeds and are fully reproducible from the released code. CMS collaboration members can reproduce the HGCAL results directly; external parties should contact the corresponding author.

8. Conclusion

We have introduced FastGraph, a PCA-subspace binned exact k-nearest- neighbor graph construction primitive for GPU geometric deep learning. The PCA-subspace formulation extends classical cell-list kNN past the small-dimensional cap that previously limited axis-aligned implementations, while a contraction argument ($V_k V_k^\top \preceq I_d$) preserves exactness in the original input space. On the HGCAL recHit workloads that motivate this paper, across d=2–10 at the headline N=5 M, k=40 operating point, FastGraph is the fastest *exact* GPU kNN method at every tested dimension, with speedups exceeding 40× over FAISS-GPU `IndexFlatL2`, 20× over cuVS brute force (the strongest exact GPU baseline), and 17× over the approximate CAGRA-NN-Descent build—which is itself the only CAGRA path

that runs at $d \geq 5$ on heterogeneous-scale detector data. The ablation in Section 5.5 isolates the PCA rotation rather than incidental implementation differences as the primary contribution: comparing each variant at its own optimum gives a within-paper ratio of 26.6× between PCA-subspace and axis-aligned binning at d=8, N=5 M, k=40.

This paper does not claim that FastGraph is the fastest GPU kNN on every distribution or every dimensionality. Two boundaries of the claim are documented. First, the deployment regime for the dynamic graph layers FastGraph targets is the d=4–10 learned latent space used by GravNet, ParticleNet, and HGCAL multi-particle reconstruction; the 3D-only spatial-acceleration methods (CLOVER, LBVH, RT-cores) are unavailable in the $d \geq 4$ regime by construction, and on d=3 HGCAL recHits the CLOVER spatio-graph index is roughly two orders of magnitude faster than FastGraph (Section 5.6). The d=3 detector-data case does not overlap with the regime FastGraph is designed for, but it delineates the boundary of the claim. Second, on *isotropic* data the PCA projection has no structure to exploit and FastGraph degrades toward ordinary cell-list behavior; for that case the relative ordering against dense BLAS brute force is determined by the cell-list pruning advantage alone, not by the PCA contribution. The HGCAL gains are an anisotropic-data result, and we expect PCA-subspace binning to be a generally useful substrate for dynamic graph layers in scientific GNNs whose latent spaces have similar spectral structure.

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Appendix A. Object Condensation Helper — Implementation Details

In addition to the kNN primitive, the library provides a GPU-resident CUDA kernel for the object condensation loss [3], widely used to cluster sensor-hit point clouds into particle candidates in physics reconstruction. Its training loop requires two dense auxiliary index matrices to be recomputed at every forward pass:

1. The *association matrix* $M \in \mathbb{Z}^{n_{\text{unique}} \times n_{\max uq}}$, whose k-th row lists the vertex indices assigned to the k-th object candidate.

2. The optional *complement matrix* $M_\neg \in \mathbb{Z}^{n_{\text{unique}} \times n_{\max rs}}$, listing vertices *not* assigned to the candidate; required only when the repulsive term $m_\neg$ is included in the loss.

Both matrices must be recomputed every forward pass because the predicted associations and the ragged batch layout change from iteration to iteration. The CUDA kernel parallelizes over unique object candidates, assigning one thread per candidate, eliminating atomic operations and thread synchronization and ensuring collision-free writes to the output matrices at the cost of $O(n_{\text{unique}})$ launched threads per forward pass.

Algorithm 2: CUDA Object Condensation Helper (`oc_helper`): one thread per candidate.

Input: `asso_idx`$[n_v]$, `unique_idx`$[n_u]$, `unique_rs_asso`$[n_u]$, `rs`$[n_s{+}1]$, $n_v, n_u, n_{\max uq}, n_{\max rs}$, flag `calc_m_not`

Output: $M[n_u][n_{\max uq}]$, $M_\neg[n_u][n_{\max rs}]$

$k \leftarrow$ `blockIdx.x` $\cdot$ `blockDim.x` + `threadIdx.x`
if $k \geq n_u$ **then**
  return
$uq \leftarrow$ `unique_idx`$[k]$; $split \leftarrow$ `unique_rs_asso`$[k]$; $start \leftarrow$ `rs`$[split]$; $end \leftarrow \min($`rs`$[split{+}1],\ n_v)$
if $end - start > n_{\max rs}$ **then**
  $end \leftarrow start + n_{\max rs}$
$fill \leftarrow 0$
for $i \leftarrow start$ **to** $end - 1$ **do**
  if `asso_idx`$[i] = uq$ **then**
    $M[fill][k] \leftarrow i$; $fill \leftarrow fill + 1$
    if $fill \geq n_{\max uq}$ **then**
      break
while $fill < n_{\max uq}$ **do**
  $M[fill][k] \leftarrow -1$; $fill \leftarrow fill + 1$
if `calc_m_not` **then**
  $fill \leftarrow 0$
  for $i \leftarrow start$ **to** $end - 1$ **do**
    if `asso_idx`$[i] \neq uq$ **then**
      $M_\neg[fill][k] \leftarrow i$; $fill \leftarrow fill + 1$
      if $fill \geq n_{\max rs}$ **then**
        break
  while $fill < n_{\max rs}$ **do**
    $M_\neg[fill][k] \leftarrow -1$; $fill \leftarrow fill + 1$